



PROJECT DESIGN PHASE



Proposed Solution

 **DATE**
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 **NAME**
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 **PROJECT**
**OptiCrop – Smart
Agricultural
Production
Optimization Engine**

 **MAX MARKS**
2 Marks

Proposed Solution Canvas

PARAMETER	DESCRIPTION
Problem statement	Farmers often select crops based on habit rather than actual soil and climate suitability, which reduces yield.
Idea or solution description	Develop a machine learning based web application that recommends the most suitable crop based on soil nutrients and climate conditions.
Novelty or uniqueness	The system provides instant, data-driven crop recommendations backed by clustering and classification analysis.
Social impact	Helps farmers make faster, more confident planting decisions and supports better use of water, fertilizer, and soil resources.
Revenue model	The solution can be offered to agri-tech companies, farming cooperatives, and extension services as a software service.
Scalability of the solution	The application can be scaled to support more crops, larger datasets, and additional regions.